Project: Worked examples
Title: Material strength margin
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

The factors of safety of a material against an applied (equivalent) stress on its yield and, when given, its ultimate strength, against a required factor; the strengths can be filled from a released material record so the margin cites its source.

WHAT IT CALCULATES	RELATION
Factor of safety against yield and against ultimate	$n_y = f_y/\sigma$; $n_u = f_u/\sigma$
Governing factor, the ultimate margin scaled to the yield basis by $k_u \geq 1$	$n_{gov} = \min(n_y, n_u/k_u)$
Utilisation against the required factor	$u = n_{req}/n_{gov}$

Basis: Design factor and factor of safety[1](§1-10).

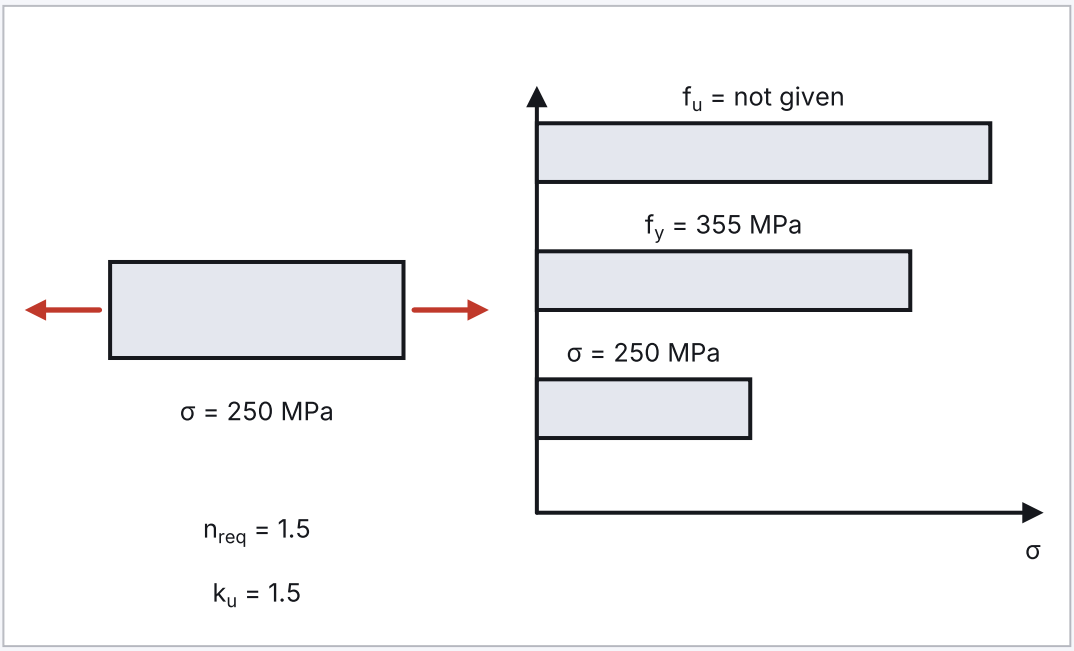


Figure 1 — A piece of the material under the applied stress, and the stresses as bars (drawn representatively, not compared). Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
σ	Applied (equivalent) stress	250	MPa	spec: LOAD-SCHEDULE-01	0.01 … 5000 MPa
f_y	Yield strength	355	MPa	drawing: DRG-BR-702 rev A	1 … 5000 MPa
k_u	Ratio of the ultimate to the yield safety factor required	1.5		spec: SPEC-MECH-03 design criteria	1 … 3

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
n_{req}	Required factor of safety on yield	1.5		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. Static loading; the applied stress is the equivalent stress at the critical point. (Fatigue and stability are separate checks.)

4. Working

[5] $n_y = f_y / \sigma$ [1](§1-10)

= 355 MPa / 250 MPa

= 1.42

Factor of safety on yield

[6] $n_{gov} = n_y$

= 1.42

= 1.42

Governing factor (yield only)

[7] $u = n_{req} / n_{gov}$

= 1.5 / 1.42

= 1.0563

Required over governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
n_y	1.42	Factor on yield
n_{gov}	1.42	Governing factor
u	1.06	Utilisation (governing)

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
FAIL	Factor of safety against the required factor	u = 1.06	u ≤ 1	1.06	[1](§1-10)

7. Applicability

None.

8. Verification

Template mech.material-margin@1.0.1 Content hash 7ddefccebe71d022 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.